Chemical Reactions and Equations

The big idea

In a **chemical reaction**, substances (reactants) change into new substances (products) with new properties. You can tell a reaction has happened from clues: a **change in state, colour or temperature**, **evolution of a gas**, or **formation of a precipitate**.

EVERYDAY REACTIONS

Milk turning sour, iron rusting, food being digested, magnesium ribbon burning with a dazzling white flame to form white magnesium oxide powder.

Writing and balancing equations

A **chemical equation** shows a reaction with symbols and formulae: $Mg + O_2 \rightarrow MgO$. This is a **skeletal** equation; it is not balanced.

LAW OF CONSERVATION OF MASS

Mass can neither be created nor destroyed in a reaction, so the number of atoms of **each element** must be the same on both sides. Balancing makes this true: **$2Mg + O_2 \rightarrow 2MgO$**.

How to balance (hit-and-trial): balance the compound with the most atoms first, then the remaining elements one at a time, never changing the formulae, only the **coefficients** in front. Add state symbols: (s), (l), (g), (aq).

BALANCE $FE + H_2O \rightarrow FE_3O_4 + H_2$

Balance O: $4H_2O$. Then H: $4H_2$. Then Fe: 3Fe.

$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

Types of reactions

Type	Pattern	Example
Combination	A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$ (slaked lime; very exothermic)
Decomposition	AB → A + B	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (heat); $2AgCl \rightarrow 2Ag + Cl_2$ (sunlight)
Displacement	A + BC → AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (blue solution turns green)
Double displacement	AB + CD → AD + CB	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$ (white precipitate)

Decomposition needs energy: **heat** (thermal), **light** (photolytic, used in black-and-white photography) or **electricity** (electrolysis of water gives H_2 and O_2 in a 2 : 1 volume ratio).

Exothermic reactions release heat (respiration, burning). **Endothermic** reactions absorb heat (decomposition of calcium carbonate).

Oxidation and reduction

REDOX

Oxidation: gain of oxygen or loss of hydrogen. **Reduction**: loss of oxygen or gain of hydrogen.

In $CuO + H_2 \rightarrow Cu + H_2O$, CuO is **reduced** to Cu and H_2 is **oxidised** to H_2O.

The substance that gets oxidised is the reducing agent.

Oxidation in daily life

- **Corrosion**: metals attacked by moisture, air and acids. Rusting of iron (reddish-brown), black coating on silver, green coating on copper.
- **Rancidity**: oxidation of fats and oils gives food a bad smell and taste. Prevented by adding antioxidants, flushing chip packets with **nitrogen**, keeping food airtight or refrigerated.

CHANGING SUBSCRIPTS WHILE BALANCING

Writing H_2O_2 instead of $2H_2O$ changes the substance. Only change the numbers **in front**.

Quick recap

- Balanced equations obey conservation of mass.
- Combination, decomposition, displacement, double displacement.
- Oxidation = gain O / lose H; reduction = lose O / gain H.
- Corrosion and rancidity are everyday oxidation.

Practice questions

1. When an iron nail is placed in copper sulphate solution, the solution changes from
   (a) green to blue (b) blue to green (c) colourless to blue (d) blue to colourless
2. The reaction $CaO + H_2O \rightarrow Ca(OH)_2$ is
   (a) a combination reaction that is exothermic (b) a decomposition reaction (c) a displacement reaction (d) an endothermic reaction
3. In $CuO + H_2 \rightarrow Cu + H_2O$, the substance oxidised is
   (a) CuO (b) H_2 (c) Cu (d) H_2O
4. Chips packets are flushed with nitrogen to prevent
   (a) corrosion (b) rancidity (c) decomposition (d) displacement
5. Electrolysis of water gives hydrogen and oxygen in the volume ratio
   (a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 3 : 1
6. Formation of a white precipitate when barium chloride is added to sodium sulphate is a
   (a) combination reaction (b) decomposition reaction (c) double displacement reaction (d) redox reaction only

Answers: 1. b 2. a 3. b 4. b 5. b 6. c